S1 Table. Data, graphs, cohorts and the four runs of the discovery-order test.

Contract-graph analysis, the main analysis of the article: approvals and transfers of WETH, USDC, USDT0, ARB and WBTC, amounts in US dollars at DefiLlama daily closing prices, every score from the twelve months before its freeze, graphs whose nodes are contracts other than smart accounts, each allowance counted up to its owner's balance at the freeze. T_{obs} is 30 June 2026 and t_1 is 31 March 2026, each at 23:59:59 UTC. The W1 scores are the same-window scores. The two score windows overlap from July 2025 to March 2026, and the W1 score window contains the W0 label window. The discovery-order test was defined after the registered results were known; it decides no rule of the analysis plan or the registration.

A. Events, account types, balances, graphs, universes, cohorts, labels and constants.

	T_{obs} (same window and W1)	t_1 (W0)
Windows		
Freeze	30 Jun 2026, 23:59:59 UTC	31 Mar 2026, 23:59:59 UTC
Score window (events behind the scores)	1 Jul 2025 – 30 Jun 2026	1 Apr 2025 – 31 Mar 2026
Label window	1 Jul 2026 – 30 Sep 2026 (W1)	1 Apr 2026 – 30 Jun 2026 (W0)
Events of the five tokens, 1 Apr 2025 to 30 Sep 2026 (both freezes)		
`Approval` rows of any value and non-zero `Transfer` rows	1,240,275,042	
with a contract at both ends	603,475,784	
Distinct addresses other than the zero address	25,537,280	
Account types (code read at block 513,216,506)		
Addresses that sent a transaction (user accounts)	17,477,779	
Addresses read with `eth_getCode`[a]	6,230,270	
no code (user account)	2,909,509	
EIP-7702 delegation designator (user account)	1,165,025	
Safe (smart account, user account)	46,204	
ERC-4337 account by the rule of the reads (smart account, user account)	87,157	
ERC-4337 account by the listed code families (smart account, user account)	192,656 (37,337 by code hash, 155,319 by code hash and implementation)	
any other code (contract)	1,829,719	
Typed user accounts / contracts, with the transaction senders	21,877,279 / 1,829,719	
Audited transaction senders with code / addresses with a call aborted at every attempt	0 of 1,000 / 0	
Balances and the balance cap		
Sampled (token, owner) pairs whose balance equals `balanceOf` at the T_{obs} block	1,493 of 1,500 (99.5%; threshold 98%)	
Positive latest allowances between contracts	339,282	308,922
unlimited	223,892 (66.0%)	208,245 (67.4%)
capped to zero (the owner holds no balance)	222,559 (65.6%)	289,299 (93.6%)
capped below the allowance	114,454	17,480
positive after the cap	116,723	19,623
sum of the capped values (USD)	68,149,271	83,952,622
Contract graph (1,403,959 contracts numbered over both freezes)		
Allowance layer: pairs (owner contracts / spender contracts)	115,006 (9,459 / 100,112)	17,940 (10,153 / 2,050)
pairs of the largest owner contract / of the largest spender contract	98,073 / 2,097	256 / 2,008
pairs without the balance cap	302,559	275,917
Transfer layer: pairs (sender contracts / recipient contracts)	2,580,709 (1,007,310 / 801,813)	2,269,073 (873,388 / 626,356)
pairs of the largest sender contract / of the largest recipient contract	130,667 / 441,612	121,708 / 439,299
Union graph: contracts with a pair / pairs	1,299,438 / 2,575,632	1,111,554 / 2,269,421
giant SCC / IN / OUT / other	265,804 / 174,239 / 242,833 / 616,562	222,592 / 169,237 / 109,804 / 609,921
Universes, importance and hot sets of the discovery-order test		
Universe of the `both` mode (WCC of the giant SCC): contracts / union pairs	1,169,044 / 2,378,418	1,015,867 / 2,121,895

	T_{obs} (same window and W1)	t_1 (W0)
capped allowance pairs / transfer pairs inside it	16,776 / 2,383,503	17,741 / 2,121,600
Universe of the `out` mode (giant SCC and OUT): contracts / union pairs	508,637 / 1,546,906	332,396 / 1,283,946
Typed contracts with a positive importance / with a positive I_A	809,296 / 52,446	715,480 / 40,395
Sum of I_T / of I_A over the typed contracts (USD)	16,611,212,142 / 309,472,584	16,622,706,250 / 343,843,976
share held by the WCC: I_T / I_A	96.5% / 93.0%	97.0% / 93.8%
share held by the giant SCC and OUT: I_T / I_A	81.3% / 90.6%	80.7% / 87.3%
Hot contracts $\lvert H_q(I_T)\rvert$ / $\lvert H_q(I_A)\rvert$, `both` mode	13,141 / 11,662	12,697 / 8,136
Hot contracts $\lvert H_q(I_T)\rvert$ / $\lvert H_q(I_A)\rvert$, `out` mode	5,087 / 5,088	3,324 / 3,324
Structure pilot at t_1: share of the giant SCC and OUT with a capped allowance in-edge	0.54% (stop threshold of the protocol 5%)	
Cohorts (contracts) and where they lie in the contract graph of their freeze		
Cohort of the freeze (≥ 3 owners on the five tokens in the score window)	6,443 of 6,483 listed (40 smart accounts are user accounts)	6,292 of 6,343 listed (51 smart accounts); 5,351 also in the matched cohort
in the allowance layer / transfer layer / union graph	1,250 (19.4%) / 4,059 (63.0%) / 4,270 (66.3%)	1,248 (19.8%) / 3,893 (61.9%) / 4,108 (65.3%)
Spender cohort	7,915 of 7,995 listed (79 user accounts, 1 system address); W1	7,864 of 7,962 listed (97 user accounts, 1 system address); W0
in the allowance layer / transfer layer / union graph	1,975 (25.0%) / 5,504 (69.5%) / 5,759 (72.8%)	1,901 (24.2%) / 5,443 (69.2%) / 5,700 (72.5%)
outside the contract graph (score zero)	2,156 (27.2%)	2,164 (27.5%)
Labels of the spender cohorts (they count every account, user accounts included)		
New approval pairs: total (share from user accounts); spenders with the label	702,275 (87.9%); 1,491	748,258 (90.4%); 1,725
New transfer senders: total (share from user accounts); spenders with the label	618,461 (82.3%); 1,369	651,474 (86.3%); 1,416
Revocations: total (share from user accounts); spenders with the label	43,625 (96.9%); 935	61,869 (97.2%); 1,113
Drained owners (heuristic): total (share from user accounts); spenders with the label	528,557 (95.3%); 1,611	570,724 (95.9%); 1,807
Constants (USD, from the W0 score window of the contract graph, two significant digits)		
Median transfer m_T / median positive capped allowance m_A	160 / 0.071	
Cap U of the amount-weighted allowance layer / cap U_{user} per user account in the importance	16,000 / 31,000	

A cell that spans both columns holds for both freezes. SCC: giant strongly connected component of the union graph; IN: contracts that reach it; OUT: contracts reached from it; WCC: the weakly connected component that holds the giant SCC. A user account is an address that sent a transaction, an address without code, an address with an EIP-7702 delegation designator or a smart account; every other typed address is a contract. I_T: US dollars that user accounts transferred to a contract in the score window; I_A: US dollars that they had authorized it to spend at the freeze, counted up to their balance; both counted up to U_{user} per account. $H_q(I)$: the contracts of a universe whose importance I is positive and among its largest, $q = 1\%$ of the universe with ties. A cohort list holds the addresses with contract code that received a non-zero approval of one of the five tokens from at least three distinct owners in the score window; a spender list holds the addresses with a positive latest allowance of one of the five tokens, set in the score window, at the freeze, among the approvals with an address of the cohort list as owner or spender. [a]The addresses that sent no transaction and have a counterparty that is neither a transaction sender nor the zero address, the members of the cohort and spender lists, and two audit samples of 1,000 addresses each.

B. $\mathrm{AUC}_{10\%}$ **of every ordering in the four runs.**

	T_{obs}, `both` (decides)		T_{obs}, `out`		t_1, `both`		t_1, `out`	
Ordering	$H_q(I_T)$	$H_q(I_A)$	$H_q(I_T)$	$H_q(I_A)$	$H_q(I_T)$	$H_q(I_A)$	$H_q(I_T)$	$H_q(I_A)$
Breadth-first (BFS)	0.0565	0.0882	0.1179	0.0595	0.0543	0.1088	0.1269	0.0595
Random	0.0609	0.0725	0.1085	0.0621	0.0543	0.0930	0.1169	0.0703
IB'_A (allowance backlinks)	0.0626	0.0881	0.1438	0.0818	0.0539	0.1094	0.1750	0.1076
IB'_T (transfer backlinks)	0.1135	0.1466	0.3023	0.1233	0.1107	0.2001	0.3460	0.1478
IB'_U (contract backlinks)	0.1136	0.1514	0.3026	0.1302	0.1276	0.2328	0.3461	0.1561
EndorseRank′	0.0621	0.0869	0.1404	0.0802	0.0526	0.1062	0.1733	0.1068
EndorseRank-activity′	0.0612	0.1042	0.1404	0.0802	0.0587	0.1426	0.1734	0.1066
AWP′	0.1465	0.1582	0.4446	0.1753	0.1524	0.2220	0.4766	0.2231
C-PR′ ($\lambda = 1$)	0.0617	0.0864	0.1398	0.0789	0.0526	0.1057	0.1728	0.1063
C-PR′ ($\lambda = 0$)	0.1061	0.0806	0.4133	0.1584	0.1088	0.1097	0.4818	0.2020
C-PR′ ($\lambda = 0.25$)	0.1081	0.0930	0.4129	0.1835	0.1110	0.1297	0.4817	0.2349
C-PR′ ($\lambda = 0.5$)	0.1085	0.0943	0.3829	0.1738	0.1118	0.1319	0.4683	0.2300
C-PR′ ($\lambda = 0.75$)	0.1091	0.0952	0.3744	0.1608	0.1121	0.1323	0.4562	0.2197
S-PR′	0.1069	0.0861	0.4190	0.1805	0.1095	0.1170	0.4874	0.2360
δ' (diagnostic, outside every comparison)	0.1643	0.1942	0.5357	0.2317	0.1775	0.2778	0.5795	0.2937
Reference: order by C-PR of the full graph	0.1061	0.1157	0.2449	0.1279	0.1165	0.1646	0.3239	0.1773
Reference: order by union in-degree of the full graph	0.1593	0.1994	0.4600	0.2142	0.1757	0.3151	0.5826	0.2846
Reference: ideal order	0.9438	0.9501	0.9500	0.9500	0.9375	0.9600	0.9500	0.9500
Reference: random curve	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500
Contracts of the universe	1,169,044		508,637		1,015,867		332,396	
Hot contracts	13,141	11,662	5,087	5,088	12,697	8,136	3,324	3,324

$\mathrm{AUC}_{10\%}$ is the mean over the hot contracts of $\max(0, 1 - t_h/\lceil 0.1|\mathcal{U}|\rceil)$, with t_h the visit at which a crawl reaches hot contract h and $\mathcal{U}$ the universe of the run, averaged over 20 seeds. In the `both` mode a visit reveals a contract's in-edges and out-edges and the universe is the WCC; in the `out` mode a visit reveals out-edges only and the universe is the giant SCC with OUT. A primed name is the count or walk computed on the contracts read so far; IB'_A, IB'_T and IB'_U count the visited contracts with an allowance edge, a transfer edge or an edge of either layer to the frontier contract. C-PR′ ($\lambda = 1$) and C-PR′ ($\lambda = 0$) are the allowance and the transfer layer of the coupled walk walked alone. The references are not crawls. The decision of the test is read in the first two columns only (Table 2 of the article, with intervals).

C. Contrasts outside the decision in the three runs that carry no verdict (items 1 and 2 of the protocol's list, and the replication contrast).

	T_{obs}, `out`		t_1, `both`		t_1, `out`	
Contrast	On $H_q(I_T)$	On $H_q(I_A)$	On $H_q(I_T)$	On $H_q(I_A)$	On $H_q(I_T)$	On $H_q(I_A)$
1. Each walk′ against the count it smooths						
EndorseRank′ − IB'_A	−0.0033 [−0.0038, −0.0029]	−0.0016 [−0.0019, −0.0013]	−0.0012 [−0.0014, −0.0011]	−0.0032 [−0.0034, −0.0030]	−0.0017 [−0.0020, −0.0014]	−0.0008 [−0.0011, −0.0006]
EndorseRank-activity′ − IB'_A	−0.0033 [−0.0038, −0.0029]	−0.0016 [−0.0019, −0.0012]	+0.0049 [+0.0035, +0.0061]	+0.0332 [+0.0300, +0.0362]	−0.0016 [−0.0020, −0.0013]	−0.0010 [−0.0012, −0.0008]
C-PR′ ($\lambda = 1$) − IB'_A	−0.0040 [−0.0044, −0.0036]	−0.0029 [−0.0032, −0.0026]	−0.0013 [−0.0014, −0.0012]	−0.0037 [−0.0039, −0.0035]	−0.0022 [−0.0025, −0.0019]	−0.0013 [−0.0015, −0.0010]
AWP′ − IB'_T	+0.1423 [+0.1289, +0.1543]	+0.0520 [+0.0424, +0.0611]	+0.0417 [+0.0381, +0.0452]	+0.0219 [+0.0157, +0.0284]	+0.1307 [+0.1167, +0.1456]	+0.0752 [+0.0624, +0.0870]
C-PR′ ($\lambda = 0$) − IB'_T	+0.1110 [+0.0992, +0.1229]	+0.0351 [+0.0258, +0.0431]	−0.0019 [−0.0056, +0.0017]	−0.0904 [−0.0971, −0.0838]	+0.1358 [+0.1216, +0.1504]	+0.0542 [+0.0418, +0.0657]
C-PR′ − IB'_A	+0.2391 [+0.2283, +0.2486]	+0.0920 [+0.0837, +0.1003]	+0.0579 [+0.0538, +0.0616]	+0.0224 [+0.0165, +0.0281]	+0.2933 [+0.2799, +0.3052]	+0.1224 [+0.1103, +0.1331]
C-PR′ − IB'_T	+0.0806 [+0.0692, +0.0922]	+0.0504 [+0.0408, +0.0596]	+0.0010 [−0.0025, +0.0046]	−0.0683 [−0.0756, −0.0609]	+0.1223 [+0.1078, +0.1363]	+0.0822 [+0.0682, +0.0937]
2. Among the walks						
C-PR′ − C-PR′ ($\lambda = 0$)	−0.0305 [−0.0343, −0.0264]	+0.0153 [+0.0108, +0.0206]	+0.0030 [+0.0021, +0.0038]	+0.0222 [+0.0195, +0.0254]	−0.0135 [−0.0182, −0.0090]	+0.0280 [+0.0213, +0.0350]
C-PR′ − AWP′	−0.0617 [−0.0709, −0.0526]	−0.0016 [−0.0091, +0.0063]	−0.0406 [−0.0442, −0.0371]	−0.0901 [−0.0967, −0.0832]	−0.0084 [−0.0182, +0.0028]	+0.0070 [−0.0029, +0.0167]
C-PR′ − C-PR′ ($\lambda = 1$)	+0.2431 [+0.2316, +0.2525]	+0.0949 [+0.0865, +0.1033]	+0.0592 [+0.0551, +0.0629]	+0.0262 [+0.0202, +0.0317]	+0.2955 [+0.2820, +0.3074]	+0.1237 [+0.1116, +0.1343]
C-PR′ − EndorseRank′	+0.2424 [+0.2312, +0.2518]	+0.0936 [+0.0852, +0.1020]	+0.0592 [+0.0551, +0.0628]	+0.0256 [+0.0197, +0.0312]	+0.2950 [+0.2816, +0.3070]	+0.1233 [+0.1112, +0.1340]
S-PR′ − C-PR′ ($\lambda = 0$)	+0.0057 [+0.0041, +0.0078]	+0.0221 [+0.0185, +0.0256]	+0.0006 [+0.0005, +0.0009]	+0.0073 [+0.0059, +0.0090]	+0.0056 [+0.0034, +0.0078]	+0.0341 [+0.0283, +0.0398]
4. Replication contrast: C-PR′ − IB'_U on the hot set of the union in-degree						
C-PR′ − IB'_U on $H_q(\text{IB}_U)$	−0.1955 [−0.2086, −0.1819]		−0.4689 [−0.4762, −0.4617]		−0.0877 [−0.1017, −0.0724]	

Differences of $\text{AUC}_{10\%}$ between two orderings with the 95% percentile interval of a paired bootstrap over the hot contracts (400 resamples, seed 42); an interval above 0 means that the first ordering reached the hot contracts sooner. C-PR′ without a value of λ is C-PR′ at $\lambda = 0.5$. $H_q(\text{IB}_U)$ is the hot set of the union in-degree of the full graph. The same contrasts of the deciding run are Table 3 of the article.

D. Descriptive readings of the deciding run (T_{obs}, `both` mode), by ordering.

	Hot contracts visited after 10% of the visits		Importance held by the contracts visited, after 1%		Importance held by the contracts visited, after 10%		Picks decided by a tie		Wasted visits in the first		Per crawl (mean over seeds)	
Ordering	$H_q(I_T)$	$H_q(I_A)$	I_T	I_A	I_T	I_A	`both`	`out`	1%	10%	Solves	Iterations
Breadth-first (BFS)	10.9%	17.8%	7.6%	6.0%	56.9%	53.5%	0.000	0.000	0.891	0.845	0	0
Random	10.0%	10.0%	6.5%	5.2%	41.4%	36.9%	0.000	0.000	0.841	0.682	0	0
IB'_A (allowance backlinks)	11.0%	15.3%	13.1%	34.5%	64.1%	68.4%	1.000	1.000	0.874	0.862	0	0
IB'_T (transfer backlinks)	16.1%	23.9%	42.1%	25.9%	74.3%	50.1%	1.000	0.999	0.678	0.888	0	0
IB'_U (contract backlinks)	16.1%	24.7%	42.1%	47.7%	74.3%	85.9%	1.000	0.999	0.675	0.888	0	0
EndorseRank′	11.0%	15.2%	13.1%	34.5%	64.0%	68.4%	0.999	0.998	0.874	0.863	540	1,322
EndorseRank-activity′	9.4%	15.4%	28.7%	51.4%	64.4%	86.5%	0.991	0.997	0.692	0.857	540	13,805
AWP′	17.6%	21.2%	80.0%	46.4%	82.0%	49.2%	0.252	0.005	0.522	0.835	542	29,042
C-PR′ ($\lambda = 1$)	10.9%	15.2%	13.1%	34.5%	64.1%	68.4%	1.000	0.999	0.874	0.863	540	1,317
C-PR′ ($\lambda = 0$)	14.0%	12.0%	70.7%	45.0%	73.1%	49.7%	0.531	0.019	0.747	0.911	540	24,303
C-PR′ ($\lambda = 0.25$)	14.1%	13.2%	71.0%	82.9%	73.3%	86.9%	0.534	0.019	0.727	0.908	540	23,328
C-PR′ ($\lambda = 0.5$)	14.4%	13.7%	71.4%	84.1%	73.5%	87.0%	0.535	0.023	0.719	0.900	540	22,594
C-PR′ ($\lambda = 0.75$)	14.6%	13.8%	71.4%	84.1%	73.5%	87.0%	0.537	0.025	0.719	0.889	540	22,273
S-PR′	14.0%	12.8%	70.7%	76.7%	73.3%	86.4%	0.531	0.019	0.743	0.910	1,080	26,024
δ' (diagnostic)	18.9%	27.3%	73.6%	87.6%	77.7%	88.9%	0.634	0.283	0.497	0.650	0	0

Hot contracts visited: the discovery curve, the share of the hot set that a crawl has visited, read at the grid point nearest 10% of the universe (mean over the 20 seeds); the random curve is at 10% there and the ideal order at 100%. Importance held: the share of the summed importance of the universe that the visited contracts hold, read at the grid points nearest 1% and 10% of the universe (mean over the seeds); the protocol reports these curves without intervals. Picks decided by a tie: the share of picks whose group of tied scores held more than one contract, in the deciding run and in the `out` mode at T_{obs}. Wasted visits: the share of the first 1% and of the first 10% of the visits that went to contracts with $I_T = I_A = 0$. Solves and power iterations: per crawl at the schedule of the deciding run ($\rho = 1/50$, about 540 recomputes of the ordering per crawl); S-PR′ solves two walks at each recompute. None of these readings enters a verdict.